

EXPERIMENT 1

IMPLEMENTATION OF CAESAR CIPHER (PYTHON):

AIM

To implement the **Caesar Cipher** substitution technique using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Read the key (shift value) from the user.
3. For each character in the plaintext:
 - If it is an uppercase letter, shift it by the key within A-Z.
 - If it is a lowercase letter, shift it by the key within a-z.
 - Leave non-alphabetic characters unchanged.
4. Display the encrypted (cipher) text.
5. Decrypt the cipher text by shifting each character backward using the same key.
6. Display the decrypted text.

Caesar Cipher in Python

CODE:

```
def encrypt(text, key):  
    cipher = ""  
    for ch in text:  
        if ch.isupper():  
            cipher += chr((ord(ch) - ord('A') + key) % 26 + ord('A'))  
        elif ch.islower():  
            cipher += chr((ord(ch) - ord('a') + key) % 26 + ord('a'))  
        else:  
            cipher += ch  
    return cipher  
  
def decrypt(cipher, key):  
    text = ""  
    for ch in cipher:
```

```

    if ch.isupper():
        text += chr((ord(ch) - ord('A') - key) % 26 + ord('A'))
    elif ch.islower():
        text += chr((ord(ch) - ord('a') - key) % 26 + ord('a'))
    else:
        text += ch

return text

plain = input("Enter the plain text: ")
key = int(input("Enter the key value: "))
cipher = encrypt(plain, key)
print("Encrypted Text:", cipher)
decrypted = decrypt(cipher, key)
print("Decrypted Text:", decrypted)

```

OUTPUT

```

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB1.py ===
Enter the plain text: HELLO
Enter the key value: 3
Encrypted Text: KHOOR
Decrypted Text: HELLO
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB1.py ===
Enter the plain text: MEET ME AFTER THE PARTY
Enter the key value: 5
Encrypted Text: RJJY RJ FKYJW YMJ UFWYD
Decrypted Text: MEET ME AFTER THE PARTY
>>>

```

RESULT

Thus, the **Caesar Cipher substitution technique** was implemented successfully using Python.